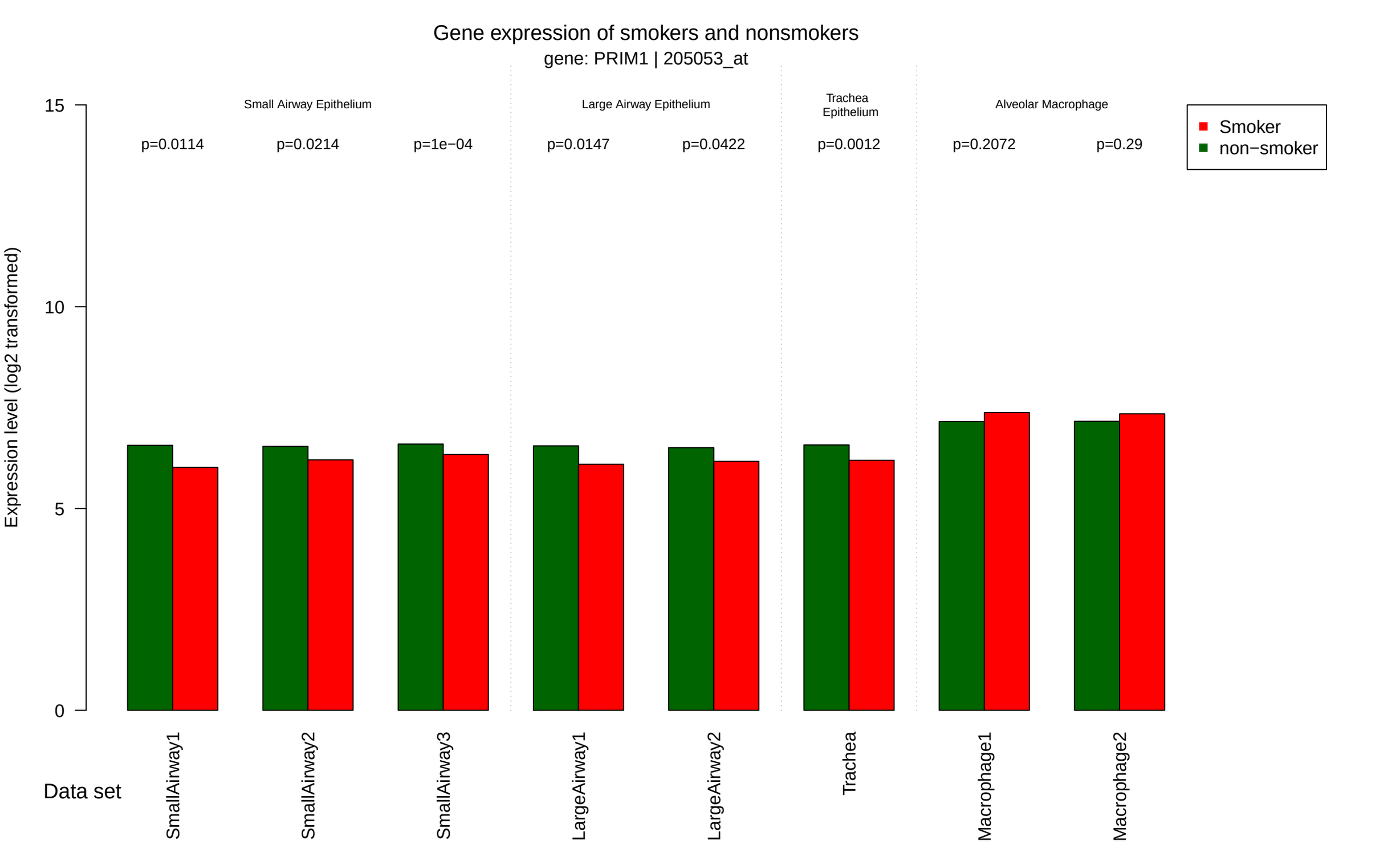
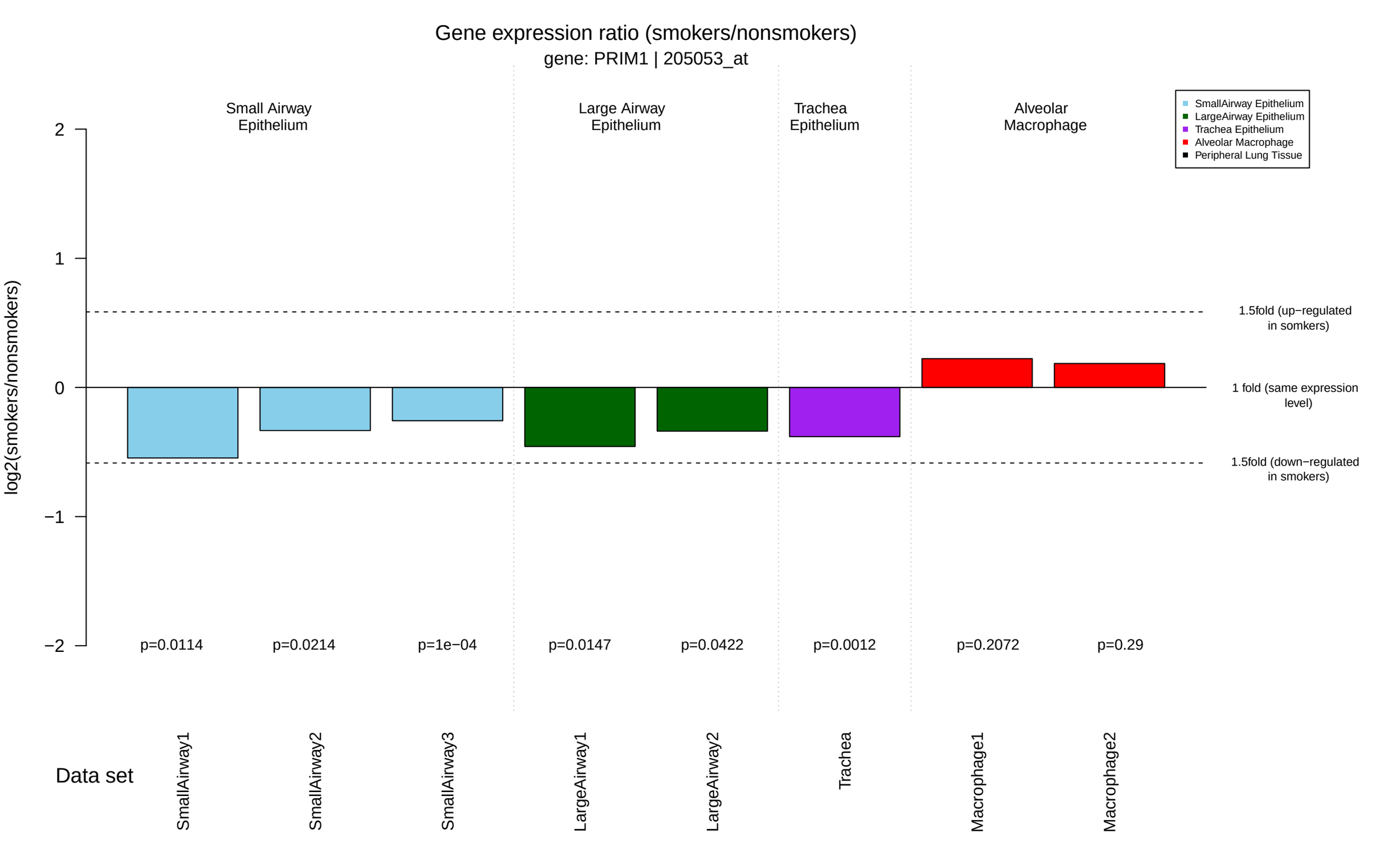
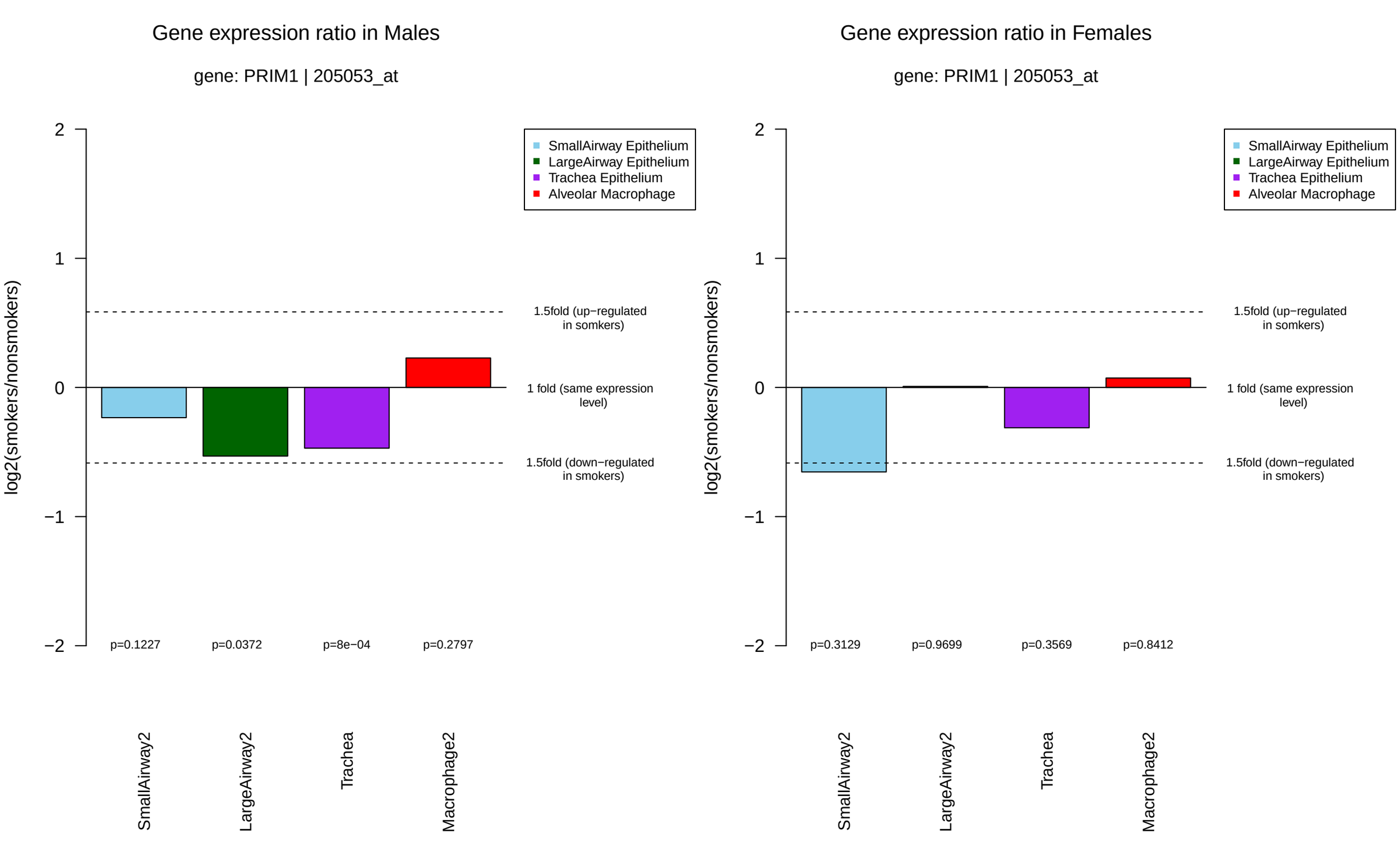
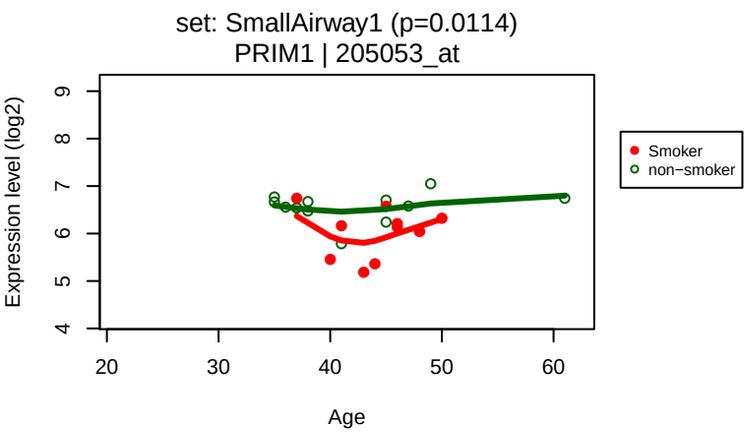
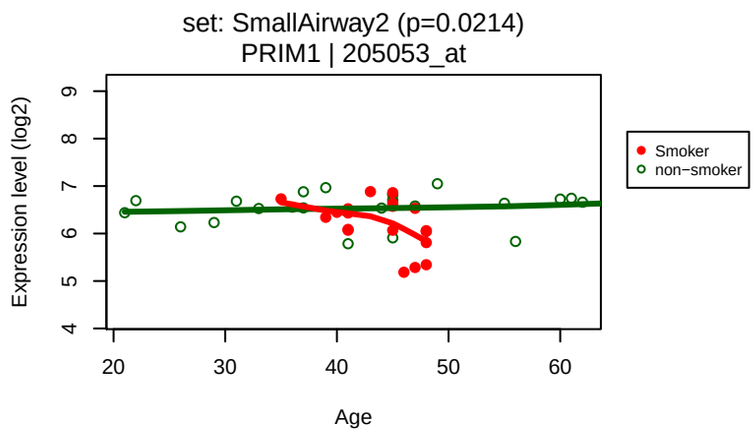
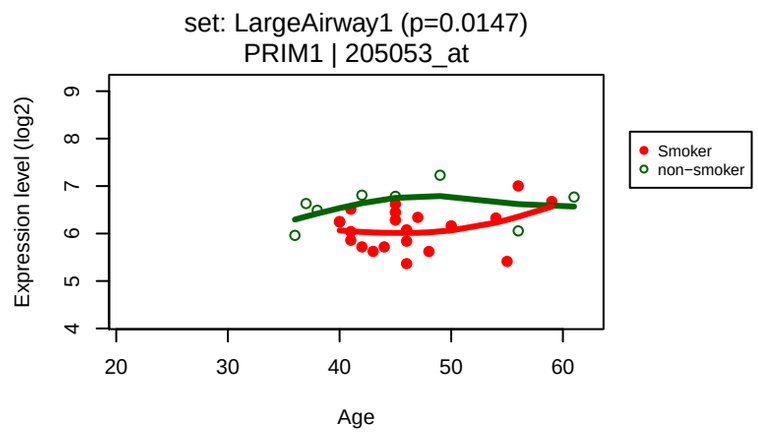

Gene expression of smokers and nonsmokers
gene: PRIM1 | 205053_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0114
p=0.0214
p=1e−04
p=0.0147
p=0.0422
p=0.0012
p=0.2072
p=0.29
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

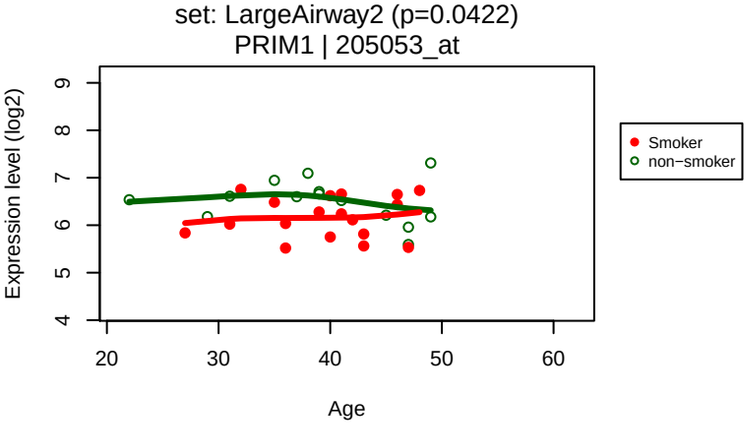

Gene expression ratio (smokers/nonsmokers)
gene: PRIM1 | 205053_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0114
p=0.0214
p=1e−04
p=0.0147
p=0.0422
p=0.0012
p=0.2072
p=0.29
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

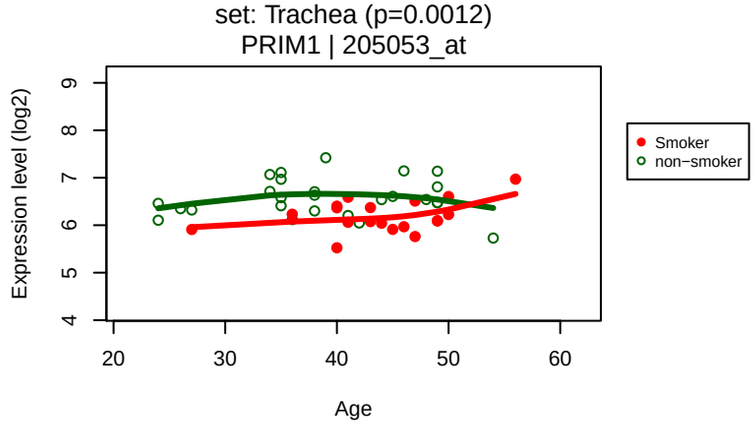

Gene expression ratio in Males
gene: PRIM1 | 205053_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1227
p=0.0372
p=8e−04
p=0.2797
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PRIM1 | 205053_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3129
p=0.9699
p=0.3569
p=0.8412
SmallAirway2
LargeAirway2
Trachea
Macrophage2

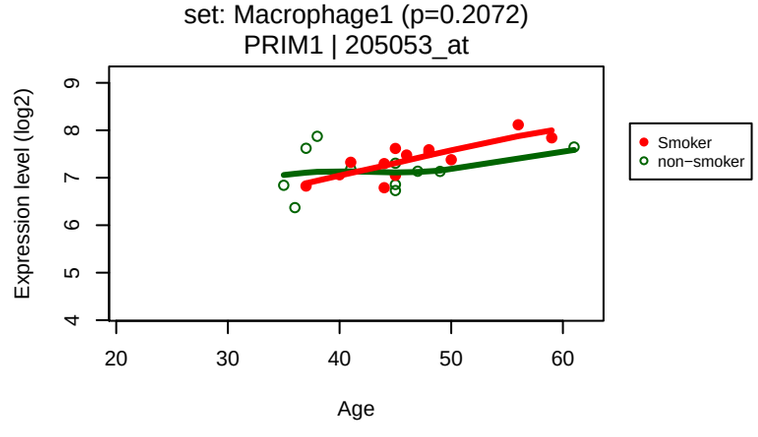

set: SmallAirway1 (p=0.0114)
PRIM1 | 205053_at
Expression level (log2)
Age
Smoker
non−smoker

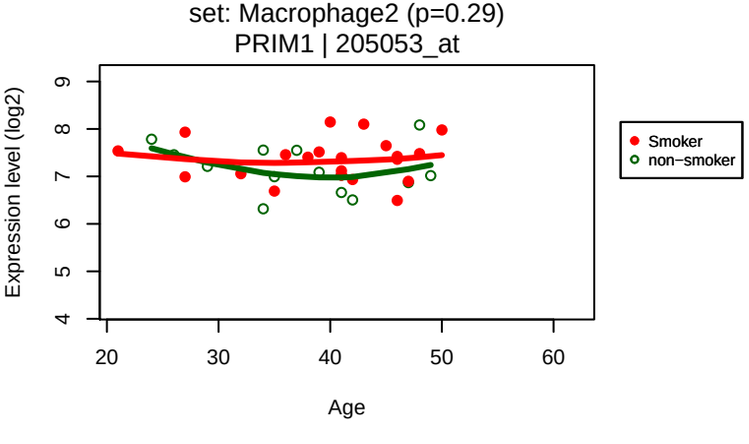

set: SmallAirway2 (p=0.0214)
PRIM1 | 205053_at
Expression level (log2)
Age
Smoker
non−smoker

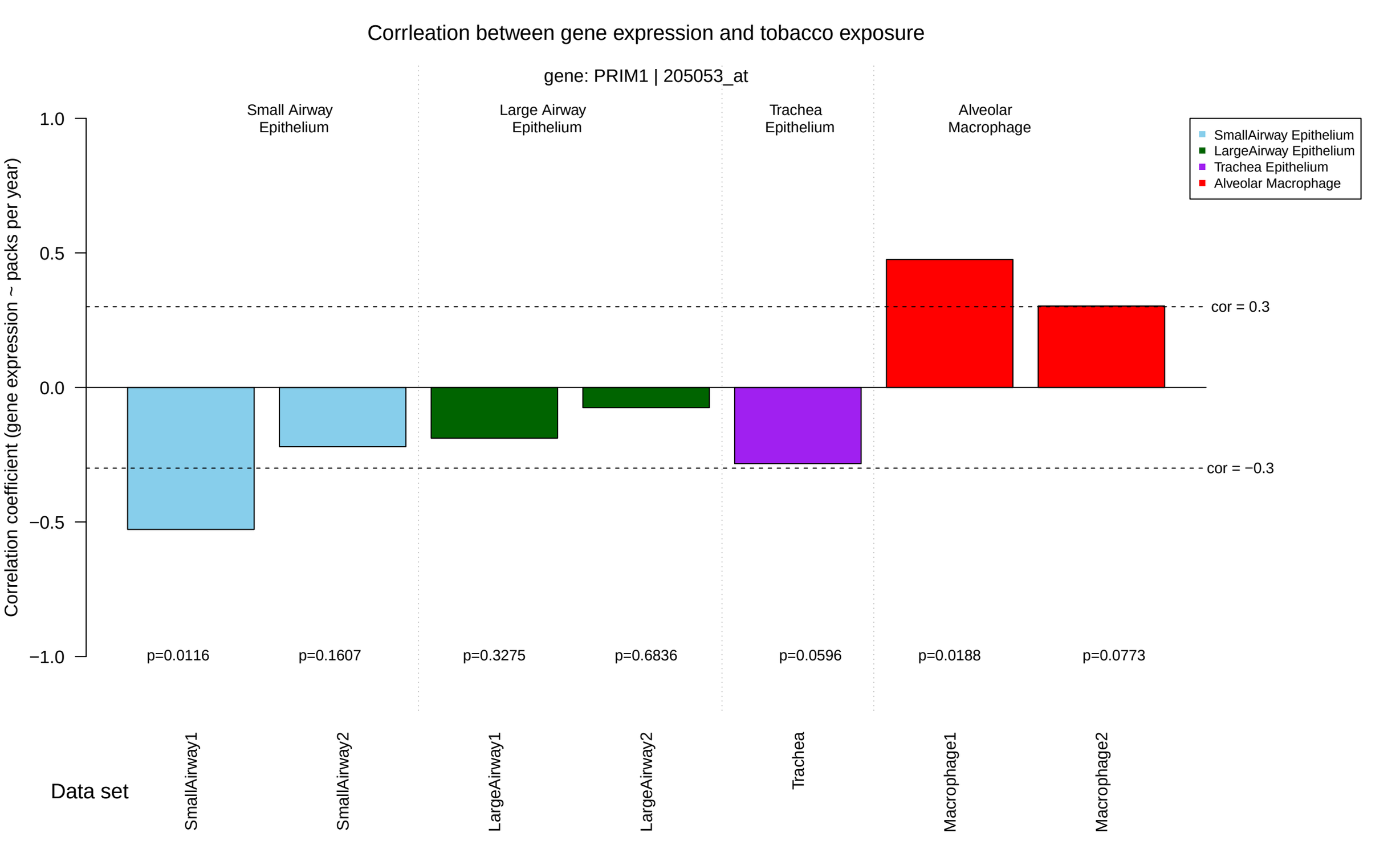

set: LargeAirway1 (p=0.0147)
PRIM1 | 205053_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0422)
PRIM1 | 205053_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0012)
PRIM1 | 205053_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.2072)
PRIM1 | 205053_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.29)
PRIM1 | 205053_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PRIM1 | 205053_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0116
p=0.1607
p=0.3275
p=0.6836
p=0.0596
p=0.0188
p=0.0773
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2